

Project: Smart NOx Sensor
Scope: Gasoline & Diesel applications
File name: 40385045_SPE_000_AJ

Handling instructions
for NOx sensors

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1 General Information

1.1 Scope of the document:

This document is a handling book listing all recommendations and rules of handling in the receiving plant and on the assembly line after reception of the sensors.

Any mishandling listed in the present document will be interpreted as a non-respect of the handling recommendations and will result in an avoidance of the warranty conditions for the parts concerned.

1.2 Analysis of the NOx Sensor

Each suspect, damaged or malfunctional NOx sensor can be sent to Vitesco Technologies GmbH for analysis to the following addresses listed below.

Please provide following information together with the failed part:

- *Customer supplier problem report with report no.*
- *Sensor upstream or downstream*
- *Latest expected response date*
- *Contact for response (name, phone, email, postal address)*
- *Vehicle type (model)*
- *Engine type*
- *Vehicle chassis no.*
- *Country of origin*
- *Mileage (km / miles)*
- *Mounting date*
- *Licensing date*
- *Removal date (repair)*
- *Part no. Customer*
- *Part identification no. from Customer (for tracking)*
- *Detailed failure description:*
 - *Temperature, environment description (engine speed, cold or hot engine, during start phase...)*
 - *Cross test results (replacement with known good SNS)*
 - *Additional defects on the affected vehicle*
 - *Diagnostic trouble codes from the dealer, vehicle tester*

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Country	Warranty Return Center	Name, First Name	Contact Info
China	Changchun	Ren, Ping (02)	Vitesco Technologies Holding China Co., Ltd 1981, Wuhan Road, Changchun 130033, Jilin Province, P. R. China Office Phone: +86 431 846 847 27 Mobile Phone: +86 130 800 379 27 Email: Ping.2.Ren@vitesco.com
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2 Handling of a pallet

2.1 Description of a pallet

If a specific pallet is in use at your receiving plant, or if a reconditioning or a repackaging happens, follow the logistics procedure and/or your internal guidelines to know how to handle pallets in your warehouse.

Use only approved packaging for transport and storage of NOx sensors

If no specific pallet is in use, Vitesco Technologies pallet and packaging are used.

- Loading unit (LU) used in Vitesco Technologies worldwide: Wooden pallet according to IPPC ISPM15 standard with a carton sleeve surrounding the primary packaging (p-pack). Typically three variants of pallets as shown in the Fig. 1, Fig. 2 and Fig. 3. are available.



Fig. 1: Standard LU
Full size



Fig. 2: Standard LU
Half size



Fig. 3: Standard LU
Quarter size

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The packaging content (e.g. for Seguin production plant) is mentioned as follows:

Characteristics	12V	24V	4p
LU Dimension[mm] Quarter size Half size Full size	630*400*580 mm 800*610*800 mm 1.200*800*930 mm	630*400*580 mm 800*610*800 mm 1.200*800*930 mm	630*400*580 mm 800*610*800 mm 1.200*800*930 mm
Gross weight LU Quarter size Half size Full size	25 kg 75 kg 174 kg	19 kg 50 kg 122 kg	24 kg 70 kg 162 kg
Number of parts / LU Quarter size Half size Full size	60 pcs 200 pcs 480 pcs	40 pcs 120 pcs 320 pcs	66 pcs 220 pcs 528 pcs
Primary packaging Dimension Content: Gross weight	560*360*130 mm 20 pc 6,5kg	535*355*175 mm 20 pc 6,5kg	560*360x130 mm 22 pc 6kg



Fig. 4: p-pack 12V



Fig. 5: p-pack 24V



Fig. 6: p-pack Gen 4P

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- Standard packaging (only Full size LU) used in Trutnov (TRU):



Fig. 7 and Fig 8: Standard loading unit in use in Trutnov

The packaging content is mentioned as follows:

Characteristics	12V	24V	4p
Dimension			
Primary packaging	600 * 400 * 118 mm	600 * 400 * 180 mm	600 * 400 * 118 mm
Loading unit	1.200*800*930 mm	1.200*800*930 mm	1.200*800*930 mm
Weight			
Loading unit	174 kg	122 kg	162 kg
Number of parts			
Primary packaging	20 pcs	20 pcs	22 pcs
Loading unit	480 pcs, 6 stacks	320 pcs, 4 stacks	528 pcs, 6 stacks

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- Standard packaging used in Changchun:



Fig. 9 and Fig 10: Standard loading unit in use for 24V in Changchun

The packaging content is mentioned as follows:

Characteristics	12V	24V
Dimension		
Primary packaging	540 * 340 * 146mm	540 * 360 * 146mm
Loading unit	1140 * 790 * 730mm	1140 * 790 * 730mm
Weight		
Loading unit	124kg	109kg
Number of parts		
Primary packaging	20pcs	15pcs
Loading unit	320pcs	240pcs

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2.2 Reception of pallet after transportation / storage

Incoming pallet

- Pallets must be inspected at arrival at customer plant and content must be checked if the tape of a packaging is broken or damaged. Acceptation with restrictions must then be notified to the transporters.
- The boxes must be stored and transported in closed condition to ensure that the anti-corrosion protection of the parts remains intact. The box should only be opened short before mounting; this means the box should be stored maximum one day at ambient temperature.
- Following steps must be ensured by the operator:
 - 1) Optical inspection of the pallet and the packaging.
 - 2) Check whether there is presence of damages on the packaging.
 - 3) Control that the pallet stands stable on the floor and that the fork-lift can operate without danger.
 - 4) Handling with care the pallet in order to avoid any shocks.

Storage / Handling of a package

- The storage environment should meet specific conditions listed in the development specification. Please refer to the internal storage instructions of NOx sensors of your company in use on the assembly line. The critical items that must be ensured in your warehouse are:

Storage temperature range	Refer to corresponding TKU / specification
Max. storage time in packaging boxes	18 months after production
Max. storage time in spare part packaging	2 years after production

- Dynamic loading allowed (Dynamic = Transit):

Quarter Size	200kg
Half Size	250kg
Full Size	350kg

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This is typically marked by the sign:



Fig.11: Quarter size marking



Fig. 12: Half size marking



Fig. 13: Full size marking

- Static loading allowed (Static = Inhouse):

Quarter Size	350kg
Half Size	430kg
Full Size	600kg

In a dynamic loading, up to 2 pallets (not more) may be stacked. Stability issue would not allow to staple more than 3 pallets in static loading. Mind the sign in Fig. 16:



Fig.14 Marking on top of the cover: „Do not weight in central“

- As specified and discussed during the development phase the handling instruction for the pallet of the NOx sensor must be followed:



Fig. 15: “Do not expose/store under rainy conditions”

- 1) The packaging must be stored in a dry and clean area. A direct exposure to water (e.g. from rain) should be limited as much as possible as shown by the sign printed on the packaging in Fig. 17. A package may therefore not be stored in rainy conditions.

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2) The orientation top/bottom is shown on the packaging by the symbol in Fig. 18:



Fig. 16: Please respect the orientation of the pallet

3) A sensor is a fragile part. A mechanical shock may damage the sensing element of the sensors as mentioned by the sign in Fig. 19:



Fig. 17: Fragile

Specific deviations due to the handling of returnable packaging must be discussed during the development phase of the project.

2.3 Error mode related to a pallet and associated actions

- Fall of a pallet:** depending on height of fall, the pallet must be sent to the test center at the assembly facility. A 100%-control must be done on the NOx sensor units as there is a high risk to use non-visible mechanically damaged parts:
 - Optical visual inspection to control any mechanical damage / shock on the part.
 - Functional control with the NOx -Tester (see Fig. 20). Each failure part must be removed and sent to the supplier.



Fig. 18: NOx Test Box

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- **Pallet with broken seal accepted without restrictions to transporters:** the responsibility of the receiving plant is engaged.
- **Parts are not transported in closed conditions:** the responsibility of the transporters is engaged.
- **Pallet stored with overload (static or dynamic):** if three or more pallets are loaded/stapled, OR if staple is done over the cross at the middle of the box, a 100%-control must be done on the NOx sensor units as there is a high risk to use non-visible mechanically damaged parts:
 - 1) Optical visual inspection to control any mechanical damage / shock on the part.
 - 2) Functional control with the NOx -Tester (see Fig. 20). Each failure part must be removed and sent to the supplier.
- **Fall of a part out of the pallet:** a fallen part **MUST** go directly to scrap.

The costs that are arising from actions listed above are not in responsibility of Vitesco Technologies.

Please handle the packaging with care

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3 Handling Rules for NOx sensor and recommendations

Please respect the following rules when handling / mounting / dismantling a NOx sensor:

3.1 When opening a box

- To open a box

- 1) Please avoid any temperature shocks when opening the shipping units. Extreme thermal shocks can be avoided by storing the parts at mounting line temperature one day before use to avoid the formation of condensation on the parts.
A sufficient re-warming of these units should be guaranteed before unpacking when the shipping units are delivered from a cold ambient condition (outside) to the in-house assembly line (at least 4 hours).
- 2) To open the packaging an appropriate tool to open the box should be used. Do not use knives. Using knives may damage the parts, especially the varnish tube.

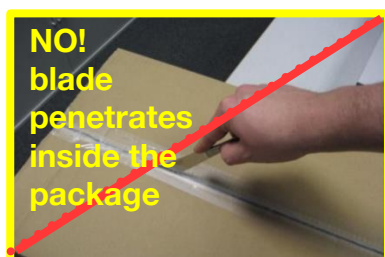


Fig. 19/20: Open the box with appropriate tool

- 3) After a package is opened, temperature variations bigger than $\pm 5^{\circ}\text{C}$ between parts and the environment must be avoided. Additionally, to avoid water droplets due to condensation, min. temperature of 15°C and max. 35°C must be considered.



Fig. 21: Strong temperature variations need to be avoided

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- 4) Maximum duration of time between cap removal and mounting of sensor probe must not be more as 5 min..
- 5) Maximum duration of time that a shipping carton can be opened before the parts are mounted, is 6 months.

• **Prevention of dirt and dust deposit when opening a box**

The unloading operation of NOx sensor out of its packaging must be done in an adequate environment: air space should be kept free from contaminations and bad weather conditions e.g. oil, water, snow, dust, sand, smoke,...

As discussed during the development phase, any deposit of dirt and dust should be prevented when assembling the sensor on the engine especially after the removal of the protection cap.

Protection caps are mainly designed to prevent dust and dirt from reaching the inside of the sensor and/ or prevent from gaseous toxics entering the probe.

They are not designed to protect the covered parts from mechanical impact.

The design of the used cap depends on the kind of shipped sensor.

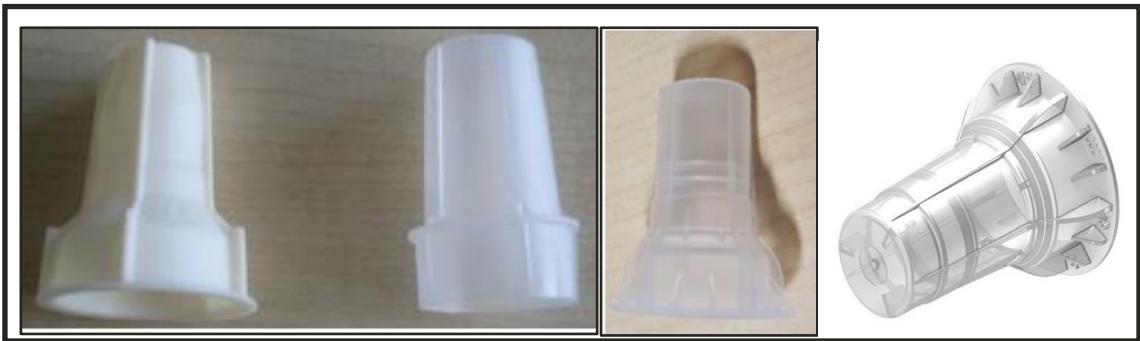


Fig. 22: Different types of protection caps

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3.2 Prevention of mishandling during assembly

Protection of the sensor during assembly/disassembly phases is very important and enhanced through following compliances:

- Gloves pertinent to ESD requirements and operator protection must be worn for assembly and disassembly phases.
- The sensor head may not be touched during assembly/disassembly phases.

The NOx Sensor is very sensitive to wrong handling in different kind. To avoid damages please consider the recommendations below:

- Mechanical shock:

For any mechanical shock on the sensor, like drop on the floor, the parts must go directly to scrap.

For other kind of mechanical shocks, in any doubt please send the NOx sensor back to the supplier for further analysis.
- Assembling of the electrical plug:
 - 1) During disassembly do not pull off the electrical plug when the NOx sensor is in use. This may cause serious damage to the engine ECU.
 - 2) Mounting: each customer determines individually, together with the project manager how to distinguish between the different sensors in the exhaust-gas system like upstream-, downstream- and OBM-sensor.

Many configurations are proposed from Vitesco Technologies:

- Different colors and different coding for connectors.

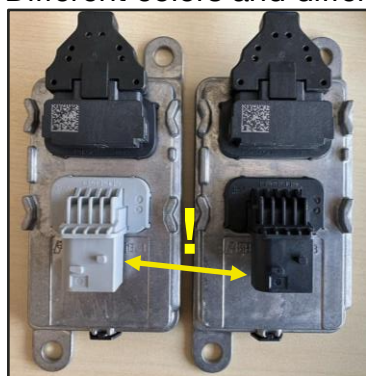


Fig. 23: Different colors of connectors

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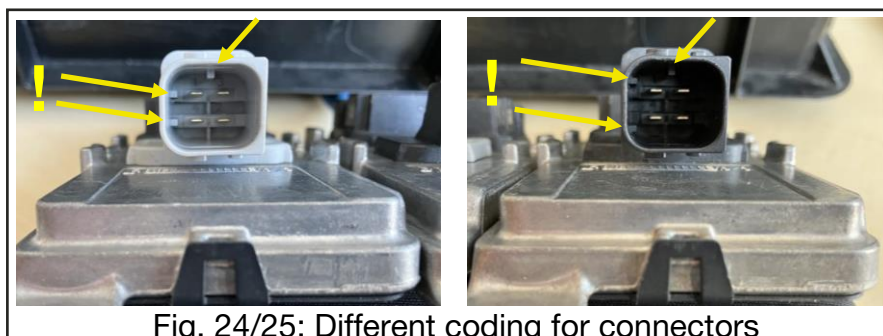


Fig. 24/25: Different coding for connectors

- Additional 5th pin for the so-called “bank detection” (e.g. if 5th pin is opened, sensor is dedicated to downstream-, when connected to ground to an upstream-position)



Fig. 26: Additional pin for “bank detection”

- Painting: Any kind of painting of the sensor or the electronic is not permitted.
- Cleaning of NOx sensor: please do not clean the sensor or the electronics mechanically or with cleaning agents. The sensor and the electrical connection may not get in contact with any fluid. This can degrade/poison the sensor or damage the label mechanically.
- Power supply: sensor plugs are equipped with mechanical Poka-Yoke preventing from a reversed assembly.
- ESD: as discussed during development phase, handling at customer must fulfill ESD standards. Vitesco Technologies suggests applying IEC 61340-5-1 and related handbook of ESD parts 61340-5-2.
- Storage on the line: store the NOx sensors on the line in an appropriate way. Do not treat them as a bulky good. Do not carry them in a bundle. Take them one by one:

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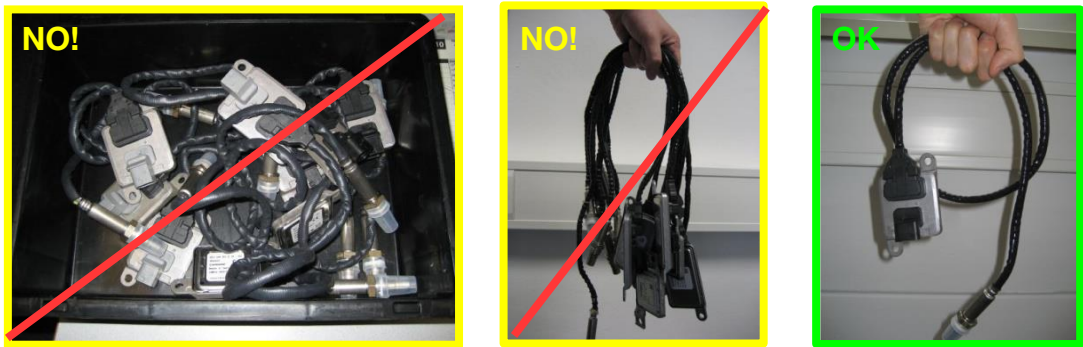


Fig. 27/28/29: Store and transport correctly the NOx sensor.

- Transport at the line: while transporting the sensor at the manufacturing line, do not leave loose the cable, because there may occur a shock on the NOx probe.



Fig. 30/31: Do not leave loose the cable

- Please mount the NOx sensor with non-damaged, dry, and clean tools. Screw tightening is allowed only with calibrated torque wrenches.

Never apply an uncalibrated tool for assembling and disassembling of a sensor

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- Make sure that no particles are inside the electrical connector before connecting
Do not try to clean it with a screwdriver or similar unsuitable tools:

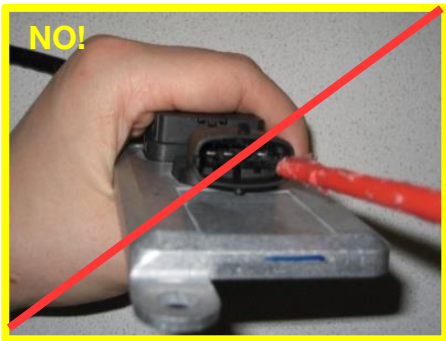


Fig. 32: Do not try to clean the electrical connector with a screwdriver or any similar unsuitable tools

- Never try to open or disconnect somehow the electrical connector to the NOx-probe:



Fig. 33: Do not try to disconnect this connector

- Do not wrap the cable tight around the electronic unit:



Fig. 34/35: Do not wrap the cable and varnish tube

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- Do not pull at the cable:

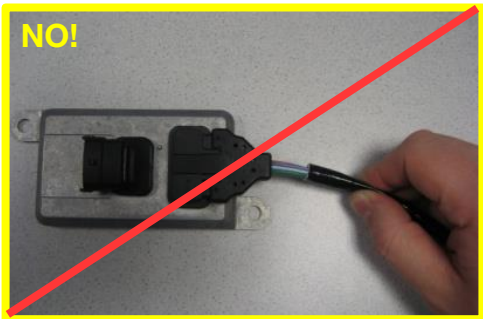


Fig. 36/37: Do not pull the varnish tube

- Do not knot the cable:



Fig. 38/39: No knot

- Do not bend the cable outgoing from the base of the sensor at the final mounting position. For transportation purpose the cable can be wrapped as shown in the picture below:



Fig. 40/41: Do not bend the varnish tube

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3.3 NOx sensor dismounting

Please be aware during handling of sensors with WEICON anti-seize paste of possible formation of hexavalent chromium (VI) in combination with chrome-nickel steel above 400 °C; The product is classified as hazardous according to Regulation (EC) 1272/2008. Refer to respective safety data sheet, provided by the supplier or SAG during acquisition and development phase. Follow the recommended safety rules and inform all people for which this information might be from importance. For upcoming changes on the material of the anti-seize paste in future, please respect the corresponding material data sheets, as well.

If it is necessary to dismount the NOx sensor, please proceed according to the following sequence:

- Check if the NOx sensor is powered off (Key off /PWL finished).
- Disconnect the electrical connector of the wiring harness.
- Unscrew the electronics (SCU).
- Unscrew the sensor part from the boss. Before unscrewing the sensor part of the boss, it may not have been powered up during 15 min as the sensor part would be very hot!

Disconnecting the sensor must be done with regards to the norm related to handling of ESD parts 61340-5-2.

The NOx Sensor is sensitive to mechanical damages or certain chemical substance. As standard tool for disassembly a spanner with narrow gap is recommended. Standard NOx sensors have a wrench size of 22 mm

- No beating with hammer is allowed.
- In case of blockage only oils without silicon or magnesium can be used. No usage of silicon substance is allowed.
- Heating up the nut also would support the disassembly. Too much heat on the sensor grommet must be avoided.
- In case that the sensor nut is fixed on the sensor, but both together can be turned out of the vehicle the electronics should be turned also to avoid twisting.
- A mounting and dismounting cycle may not occur more than 2 times for one given sensor probe.

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3.4 Prevention of mishandling during disassembly

Refer to the chapter “3.2 Prevention of mishandling during assembly”.

3.5 Re-Assembly

While reusing a sensor, the handling and mounting conditions must be ensured as defined before in the chapter 3.2.

Please note:

Only 2 sensor mountings and up to 20 electrical connections / disconnections are allowed

3.6 Prevention of chemical poisoning of the sensor during assembly

The NOx sensor is very sensitive to the chemical species Magnesium (Mg) and Silicon (Si). A shift of the characteristics can follow when a part is poisoned. A funded analysis of the presence of the species Mg and Si in the target application is strongly recommended:

- Si- or Mg-based sprays -contained e.g. in some lubricants or the thread paste may not be used.
- Only Magnesium and Silicon free material may be used for the exhaust pipe in order to prevent contamination.

Vitesco Technologies will refuse warranty if damages and failures are consecutive to a non-respect of these basic recommendations

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